If you are studying physics in high school then refer *Physics* 5th edition by Halliday, Resnick, Krane. This guide is intended for those who want to complete a core education in physics. I am assuming you have mathematical knowledge up until mentioned in this PDF. We start with introductory physics and then work our way through the undergraduate core as well as electives you may take.

1. Mathematics that you will encounter throughout your undergraduate degree.
    - *Mathematical Methods in the Physical Sciences* by Boas
    - *Mathematical Methods for Physicists* by Arfken, Weber, Harris
2. We go through basic mechanics, electrostatics and magnetostatics and modern physics.
    - *Fundamental Laws of Mechanics* and *Basic Laws of Electromagnetism* by Irodov
    - *Introduction to Mechanics* by Kleppner, Kolenkow and *Introduction to Classical Mechanics* by Morin
    - *Electricity and Magnetism* by Purcell, Morin [supplement with *Div, Grad, Curl, and All That* by Schey]
    - *Vibrations and Waves* by French and *Waves* by Crawford
    - *Introduction to Quantum Mechanics* by Griffiths, Schroeter
3. We go through the undergraduate core of a physics degree.
    - *Classical Mechanics* by Taylor
    - *Introduction to Electrodynamics* by Griffiths [supplement with *Notes on Vector Calculus* by Tong]
    - *Thermal Physics* by Schroeder or *Fundamentals of Statistical and Thermal Physics* by Reif
    - *Optics* by Hecht
    - *Principles of Quantum Mechanics* by Shankar and *A Modern Approach to Quantum Mechanics* by Townsend
4. Mathematics that you will encounter throughout your graduate years and beyond.
    - *Mathematical Physics: A Modern Introduction to Its Foundations* by Hassani
    - *Mathematical Methods of Classical Mechanics* by Arnold
    - *Advanced Mathematical Methods for Scientists and Engineers* by Bender, Orszag
5. The graduate core of a physics PHD.
    - *Modern Classical Mechanics* by Helliwell, Sahakian and *Classical Mechanics* by Goldstein, Poole
    - *Modern Electrodynamics* by Zangwill and *Classical Electrodynamics* by Jackson
    - *Statistical Physics* by Kardar
    - *Quantum Mechanics Lecture Notes* by Littlejohn and *Lecture Notes on Topics in Quantum Mechanics* by Tong

Try to find lecture notes on all topics you are studying of appropriate level such as David Tong's Lecture Notes and equivalent. Advice on undergraduate electives as well as graduate electives in physics can be found here in this post. One can only learn physics by solving problems so solve a lot and try to understand. For PHD qualifying / graduate qualifying, exams these texts are quite good to go through.

- *Problems in General Physics* by Irodov
- *Problems and Solutions on Mechanics, Electromagnetism, Quantum Mechanics, Thermodynamics and Statistical Mechanics, Optics, Atomic/Nuclear/Particle Physics, Solid State Physics, and Relativity and Cosmology* by Lim Yung-Kuo
- *Princeton Problems in Physics with Solutions* by Newbury, Newman, Ruhl, Staggs, Thorsett
- *A Guide to Physics Problems* by Cahn, Nadgorny
- The online qualifying exam archives of Princeton, Berkeley, Stanford, MIT, Chicago and Harvard, which are free and have the real exams.

A few books that have really good problems are listed below:

- *Analytical Dynamics of Particles and Rigid Bodies* by Whittaker
- *Static and Dynamic Electricity* by Smythe
- *Problems in Classical Electromagnetism* by Cherny, Kuz'mich / *Problems in Electrodynamics* by Batygin, Toptygin
- *Collection of Problems in Classical Mechanics* by Kotkin, Serbo
- *Problems in Mathematical Physics* by Budak, Samarskii, Tikhonov
- *Exploring Quantum Mechanics* by Galitski, Karnakov, Kogan, Galitski.
- *Selected Problems in Quantum Mechanics* by ter Haar (from Gol'dman, Krivchenkov).
- *Statistical Mechanics: An Advanced Course* with Problems and Solutions by Kubo.